Assignment: Hyper-Personalised Gift Recommendation Agent

Objective

Build an AI workflow that takes enriched professional contact data and recommends the **top 3 personalised gifts** for each contact.

The recommendations should be based on real signals from LinkedIn-style profile data and should include **real purchasable product links** from the web.

We want to evaluate your ability to build reliable AI product workflows — not just prompts or chatbot demos.

Problem Statement

Given a list of contacts enriched with LinkedIn-scraped content, build an AI agent/workflow that:

1. Understands each contact's professional profile, interests, recent posts, comments, and engagement.
2. Extracts safe and relevant gifting signals.
3. Searches the web using any free search API or public search approach.
4. Finds real purchasable gifts that match the contact, occasion, budget, and country.
5. Ranks the best 3 gift options.
6. Explains the reasoning behind each recommendation.
7. Generates a short personalised message to send along with the gift.
8. Supports a human-review step before the recommendations are finalised.

Sample Input


```
[
  {
    "name": "Aarav Mehta",
    "role": "VP Sales",
    "company": "Acme Corp",
    "location": "Bengaluru, India",
    "linkedin_profile": {
      "headline": "VP Sales at Acme Corp | Enterprise SaaS | GTM Leadership",
      "about": "I enjoy building high-performing revenue teams and scaling
SaaS businesses across India and Southeast Asia.",
```

```
      "experience": [
        {
          "title": "VP Sales",
          "company": "Acme Corp",
          "description": "Leading enterprise sales, strategic accounts, and GTM expansion."
        },
        {
          "title": "Regional Sales Director",
          "company": "Salesforce",
          "description": "Managed enterprise accounts across BFSI and retail."
        }
      ],
      "recent_posts": [
        "Great sales teams are built on trust, coaching, and consistency.",
        "India’s SaaS ecosystem is entering an exciting new phase.",
        "Still recovering from yesterday’s India vs Australia match. What a game!"
      ],
      "recent_comments": [
        "Completely agree — pipeline discipline matters more than heroic end-of-quarter selling.",
        "Loved this take on customer-led growth.",
        "Cricket teaches leadership better than most management books."
      ],
      "engaged_topics": [
        "Cricket",
        "Revenue leadership",
        "SaaS GTM",
        "Customer-led growth",
        "Team building"
      ]
    },
    "relationship_context": {
      "relationship_type": "Prospective customer",
      "last_interaction": "Positive discovery call last week",
      "business_goal": "Nurture relationship before follow-up meeting"
    },
    "gift_context": {
      "occasion": "Post-meeting thank you",
      "budget_min": 3000,
      "budget_max": 5000,
      "currency": "INR",
      "country": "India"
    }
  }
]
```

Expected Output

For each contact, return structured output similar to:

```
{
```

```
  "contact_name": "Aarav Mehta",
  "profile_signals": {
    "strong_signals": [
      "Interested in cricket",
      "Works in SaaS sales leadership",
      "Engages with GTM and revenue leadership content"
    ],
    "weak_signals": [
      "May appreciate leadership or business books"
    ],
    "signals_to_avoid": [
      "Do not infer religion, politics, health, family status, or other sensitive personal attributes"
    ]
  },
  "search_trace": {
    "queries_used": [
      "premium cricket gift for business professional India under 5000",
      "sales leadership gift India under 5000"
    ],
    "products_considered_count": 8
  },
  "recommended_gifts": [
    {
      "rank": 1,
      "gift_name": "Example product name",
      "product_url": "<https://example.com/product>",
      "store": "Amazon / Flipkart / other",
      "estimated_price": "₹3,999",
      "why_this_gift": "Clear explanation of why this gift matches the contact.",
      "personalisation_reasoning": "Specific profile signals used to recommend this gift.",
      "personalised_message": "A short, warm, professional note that can be sent with the gift.",
      "confidence_score": 0.87,
      "risk_level": "low",
      "assumptions": [
        "Interest in cricket inferred from recent posts and comments"
      ]
    }
  ],
  "human_review": {
    "status": "pending_review",
    "available_actions": ["approve", "reject", "edit", "regenerate"]
  }
}
```

Workflow Expectations

Your solution should be implemented as a **multi-step AI workflow**, not a single prompt.

At minimum, your workflow should include:

1. Contact profile ingestion
2. Signal extraction from profile, posts, comments, experience, and engagement topics
3. Gift-safe signal filtering
4. Product search using a free search API or public search approach
5. Product validation for relevance, price, availability, country, and appropriateness
6. Gift ranking
7. Personalised message generation
8. Human review state where recommendations can be approved, rejected, edited, or regenerated

Please show intermediate outputs for at least:

- Extracted profile signals
- Search queries used
- Products considered
- Final ranking
- Assumptions and confidence

Requirements

Your solution should include:

- Working prototype
- Support for multiple contacts
- Backend API or runnable workflow
- Structured input and output
- Real gift recommendations with valid product links
- Ranking logic for top 3 gifts
- Personalised note generation
- Human-review step
- Basic validation and fallback handling
- Clear README with setup and usage instructions

You may use any free search API or public search approach to find products. The gifts should not be invented by the model.

Guardrails

The system should avoid:

- Recommendations based on religion, politics, health, ethnicity, gender, family status, or other sensitive personal attributes
- Overly personal or creepy gift suggestions
- Gifts that are inappropriate for a professional relationship
- Hallucinated products or fake URLs
- Claims that are not supported by the available profile data

When context is weak, the system should lower confidence, state assumptions, or ask for human input instead of pretending certainty.

Suggested Tech Stack

You may use any stack you are comfortable with.

Recommended options:

- Python
- FastAPI or similar backend framework
- LangGraph, LangChain, DeepAgents, or equivalent workflow framework
- OpenAI / Anthropic / Gemini / open-source models
- Any free search API or public product search approach
- React / Next.js for UI
- CopilotKit for agent UX, if useful
- Postgres / SQLite / Redis, if useful

The stack is not mandatory. We care more about your product thinking, workflow design, and implementation quality.

Evaluation Expectations

Include a short evaluation note covering how you would measure:

- Gift relevance to contact profile
- Validity of product links
- Budget and country fit
- Professional appropriateness
- Avoidance of sensitive or creepy personalisation
- Quality of personalised message
- Failure handling when search results are poor

You do not need to build a full eval system, but we want to see how you think about quality.

What We Will Evaluate

We will evaluate your submission on:

1. **AI workflow design**

   How well you break the problem into extraction, search, ranking, validation, generation, and review.

2. **Quality of recommendations**

   Whether the gifts are thoughtful, relevant, professional, and actually purchasable.

3. **Grounding**

   Whether product links are real and fit the given budget/location constraints.

4. **Reasoning quality**

   Whether the agent explains recommendations clearly and avoids hallucinated reasoning.

5. **Guardrails**

   Whether the system avoids sensitive, risky, or creepy personalisation.

6. **Human-in-the-loop design**

   Whether a user can review, edit, approve, reject, or regenerate recommendations.

7. **Code quality**

   Clean architecture, readable code, structured outputs, and good setup instructions.

Bonus Points

Bonus points for:

- Clean LangGraph or DeepAgents-style workflow design
- UI for reviewing and editing recommendations
- Search result validation
- Retry/fallback flow when search results are poor

- Agent traces or intermediate reasoning summaries
- Basic eval examples or test cases
- Bulk upload flow
- Support for different tones in personalised messages
- Cost, latency, or tool-call logging

Submission

Please submit:

- GitHub repository
- README with setup instructions
- Sample input and output
- Short architecture note
- Demo video or screenshots, if available
- Notes on tradeoffs and future improvements

Final Note

We are less interested in whether you can call an LLM API, and more interested in whether you can design an AI system that produces useful, trustworthy, and reviewable recommendations from imperfect real-world data.